

The Primitive Generalized Fermat Equation

$$x^3 + y^5 = z^7$$

From the published frontier to a certified proof

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Abstract

We prove that the generalized Fermat equation

$$x^3 + y^5 = z^7$$

has no solution in nonzero coprime integers, and we make explicit how the proof grows from the previously published frontier. Dahmen–Siksek had established the signed local descent and eliminated the three cases in which the associated degree-seven algebra is reducible. Putz had then proved that the remaining irreducible case can involve only seven septic fields: six pure fields and one exceptional field. The missing step was to exclude those seven fields.

For the six pure fields we construct an explicit Fano resolvent, descend to a smooth plane quartic of genus three over $\mathbf{Q}(\sqrt{-7})$, and prove that its rational-parameter locus consists only of five points above the branch values. For the exceptional field we combine the modularity and conductor results of Pacetti–Villagra Torcomian with level lowering over $\mathbf{Q}(\sqrt{5})$ and a finite Hecke comparison at 29. We also reconstruct the three published reducible-sector arguments as independently replayable calculations, adding a database-free identification of the quadratic field and an intrinsic formal-group treatment at the ramified prime. The paper labels every major step as published input, reconstructed input, or new argument. All project-specific finite calculations are supplied as exact certificates, programs, inputs, and authenticated logs.

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1 Introduction: the frontier and the completion

1.1 The equation and the theorem

For pairwise coprime nonzero integers, the equation

$$x^3 + y^5 = z^7 \tag{1}$$

is the generalized Fermat equation of signature $(3, 5, 7)$. We shall use the variable order

$$(X, Y, Z) = (y, x, z), \quad X^5 + Y^3 = Z^7. \tag{2}$$

All signs are allowed. Since a prime dividing two of X, Y, Z also divides the third, primitivity is equivalent here to pairwise coprimality. Our main result is the following.

Theorem 1.1. *There are no nonzero primitive integers x, y, z satisfying*

$$x^3 + y^5 = z^7.$$

The purpose of this introduction is not only to state the theorem, but to say precisely what was known before the present work and what had still to be done. This distinction matters in a long computer-assisted proof: it makes clear which general theorems are imported, which earlier computations are reconstructed for confidence, and which arguments close the remaining gap.

1.2 The published frontier

The mathematical starting point was already far advanced. Dahmen and Siksek proved the following strong partial result [1, Theorem 1]: every nontrivial primitive solution of $X^5 + Y^3 = Z^7$ satisfies one of the alternatives

$$2 \cdot 3 \cdot 5 \mid Z \text{ and } 7 \nmid XY, \quad \text{or} \quad 2 \nmid Z, 3 \nmid XYZ, 5 \nmid YZ, 7 \mid Y. \tag{3}$$

In particular, a primitive solution for which $3 \mid XY$, $5 \mid Y$, or $7 \mid X$ is trivial. Their descent introduces the degree-seven algebra used below, proves that only four factor-degree patterns can occur, and eliminates the three reducible patterns

$$[1, 6], \quad [2, 5], \quad [3, 4].$$

The only surviving pattern is therefore the irreducible one, [7].

Putz then made this residual problem finite. His exhaustive Hunter enumeration, with the prescribed local algebra data, proves that an irreducible descent algebra must be one of exactly seven fields [2, Theorem 3.50]. Six are pure septic fields and the seventh is an exceptional septic field. Thus the published frontier can be summarized in one sentence:

a nontrivial primitive solution would have to pass through one of seven explicit irreducible degree-seven fields.

This is a remarkably narrow frontier, but it is not yet a contradiction. One must still connect each field to an arithmetic object on which rational points or residual Galois representations can be exhausted.

1.3 What is added here

The principal new work is the elimination of those seven fields.

- (i) For the six pure fields we construct the missing geometric bridge. The thirty Fano planes attached to the septic cover split into two A_7 -orbits. Their compatible pairs give a degree-120 resolvent whose quotient is an explicit smooth plane quartic C_+ over $\mathbf{Q}(\sqrt{-7})$. A rational-parameter theorem shows that the relevant parameter on C_+ takes rational values only at five points above 0, 1, and ∞ . A nonzero primitive solution would have parameter outside that set, so all six pure fields are excluded.
- (ii) For the exceptional field we place the specialization in the Frey family of Pacetti–Villagra Torcomian, use their modularity and conductor theorems together with Hilbert level lowering, and reduce both the irreducible and reducible residual alternatives to finite lists. At a prime above 29, the four curve trace polynomials and the six possible modular or character factors have 24 nonzero resultants modulo 7. This excludes the exceptional field.
- (iii) We reconstruct the previously published reducible sectors in an exact certificate framework. This is not presented as a new elimination of those sectors. It provides an independent route through the finite arithmetic and adds two genuine refinements: the quadratic factor is identified from its intrinsic local labels without a field database, and the ramified 7-adic step is carried out inside the formal neighbourhood of the given integral Jacobian model.
- (iv) We compose all four sectors into a single machine-checkable theorem graph. Every project-specific finite assertion is tied to exact input bytes, an explicit predicate, and a replay program or authenticated completed log. Published general theorems remain ordinary bibliographic dependencies.

1.4 A provenance-aware proof map

The following table is the shortest guide to the logical ownership of the proof. “Reconstructed” means that this paper supplies an independent exact certificate for a result whose mathematical argument was already in the literature; it does not claim priority for that result.

stage	status before this paper	role in this paper
signed local descent and four factor sectors	published by Dahmen–Siksek	imported, restated, and independently checked
sectors [1, 6], [2, 5], [3, 4]	eliminated by Dahmen–Siksek	reconstructed with certificates; two local refinements are added
irreducible sector [7]	reduced by Putz to seven explicit fields	the seven-field list is imported and authenticated
six pure septic fields	not eliminated at the published frontier	excluded here by the Fano resolvent and the rational-parameter theorem
one exceptional septic field	modular tools available, but no terminal comparison	excluded here by level lowering, ray characters, and the $q = 29$ resultant test
global theorem	one finite irreducible frontier remained	all four sectors are composed into the contradiction of Theorem 1.1

1.5 How the proof was completed

The working principle was to treat a hypothetical solution not as a large integer triple to be chased, but as one mathematical state seen through several exact representations. The same state has a descent algebra, local factor labels, ideal classes, divisor classes, finite-quotient coordinates, and p -adic analytic coordinates. Each representation supplies a constraint. The proof succeeds when the intersection of all compatible states is empty.

This viewpoint dictated a calm order of work:

1. fix the strongest published frontier before beginning a new search;
2. translate each remaining assertion into a finite interface with exact inputs and a stated acceptance predicate;
3. use exploratory computation only to discover a model, prime, or quotient, and then replace discovery by a deterministic certificate;
4. keep the four descent sectors separate until each has a terminal statement, and only then compose them;
5. distinguish a recomputation of a published argument from a genuinely new bridge or obstruction.

This is why the proof is modular. A specialist can inspect the new irreducible argument without repeating the reducible sectors, or can replay one finite computation without trusting the remainder of the software tree.

The central object throughout is the polynomial

$$\Phi(T) = 15T^7 - 35T^6 + 21T^5, \quad \eta = \frac{X^5}{Z^7}, \quad f_\eta(T) = \Phi(T) - \eta. \quad (4)$$

The associated étale algebra

$$A_\eta = \mathbf{Q}[T]/(f_\eta)$$

encodes a hypothetical solution in several complementary forms. Local factorization describes which global fields may occur. Descent maps those fields to curves. Mordell–Weil information restricts the resulting points to finitely many congruence classes, and local analytic functions decide whether those classes contain a point compatible with the original rational parameter.

The argument uses standard published results in descent, number-field enumeration, modularity, level lowering, formal groups, Mordell–Weil sieves, and Chabauty methods. The large finite

calculations specific to (1) are exact and are separated into independently checkable certificates. Thus the proof has two visible layers: general theorems, cited in the usual way, and finite arithmetic, supplied with the paper. Sections 2–5 reconstruct the descent and reducible sectors. Section 6 begins at Putz’s seven-field frontier and contains the new terminal argument. The final sections compose the proof and describe the certificates.

2 The degree-seven descent algebra

Provenance. The local covering and exhaustive four-sector reduction in this section are published inputs from Dahmen–Siksek. We record them because they are the interface between a primitive solution and every later computation.

The derivative and the two branch values are given by

$$\Phi'(T) = 105T^4(T-1)^2, \quad \Phi(0) = 0, \quad \Phi(1) = 1. \quad (5)$$

In particular, Φ is strictly increasing on $\mathbb{R}$. For a nonzero solution, $\eta \notin \{0, 1\}$, and f_η is separable with one real root. A resultant computation gives

$$\text{disc}(f_\eta) = -3^6 5^6 7^7 \eta^4 (\eta - 1)^2. \quad (6)$$

Its squareclass is therefore -7 .

The local covering theorem at 7 in [1, Proposition 3.5] gives local factor-degree patterns $[1, 6]$, $[2, 5]$, $[3, 4]$, and $[7]$. Global factors induce groupings of the irreducible local factors, so their degrees are obtained by coarsening these patterns. Consequently the possible global factor-degree patterns are exactly among

$$[1, 6], \quad [2, 5], \quad [3, 4], \quad [7]. \quad (7)$$

In particular, there are at most two global factors. For example, if $7 \mid X$ and $r = \text{ord}_7(X)$, the Newton polygon has vertices $(0, 5r)$, $(5, 1)$, $(7, 0)$, giving irreducible local factors of degrees 5 and 2. If $7 \mid Y$ and $r = \text{ord}_7(Y)$, translation by 1 gives vertices $(0, 3r)$, $(3, 1)$, $(7, 0)$, giving irreducible local factors of degrees 3 and 4. The unit case, and the $7 \mid Z$ case after scaling to an integral model, give the remaining local patterns by the cited proposition.

We now eliminate the four sectors one at a time.

3 The rational-factor sector

Provenance. The mathematical elimination in this section is due to Dahmen–Siksek. This section gives a certificate-level reconstruction of their Proposition 6.1 and Lemmas 6.2–6.8 [1, Section 6].

Assume that $r \in \mathbf{Q}$ is a root of f_η , and put $w = rZ$. Clearing denominators gives

$$15w^7 - 35Zw^6 + 21Z^2w^5 - X^5 = 0. \quad (8)$$

The translated identity

$$\Phi(T) - 1 = (T-1)^3(15T^4 + 10T^3 + 6T^2 + 3T + 1)$$

also gives

$$(w - Z)^3(Z^4 + 3Z^3w + 6Z^2w^2 + 10Zw^3 + 15w^4) = -Y^3. \quad (9)$$

3.1 Local restrictions and quadratic ideal descent

Newton polygons at 3, 5, and 7 give

$$3 \nmid Z, \quad 5 \nmid YZ, \quad 7 \nmid Y, \quad \text{ord}_3(r) \in \{0, 1, -1\}, \quad \text{ord}_5(r) \geq 0. \quad (10)$$

The rational-root theorem applied to (8) then shows that w is an integer or one third of an integer. Moreover, for every prime p , at least one of $\text{ord}_p(Z)$ and $\text{ord}_p(w)$ is zero.

For all three reducible sectors fix the notation

$$k = \mathbf{Q}(\omega) = \mathbf{Q}(\sqrt{-35}), \quad \omega^2 + \omega + 9 = 0, \quad \delta = 2\omega + 1 = \sqrt{-35}. \quad (11)$$

Thus $\mathcal{O}_k = \mathbf{Z}[\omega]$. Put

$$A = Z^2 + wZ - 3w^2, \quad B = -wZ - w^2, \quad \mathcal{F} = A + B\omega.$$

Then

$$\mathcal{F}\overline{\mathcal{F}} = A^2 - AB + 9B^2 = Z^4 + 3Z^3w + 6Z^2w^2 + 10Zw^3 + 15w^4.$$

The class group of k is C_2 . The prime 3 splits, and exact ideal arithmetic reduces (9) to three possibilities:

$$\mathcal{F} = \alpha(u + v\omega)^3, \quad \alpha \in \{1, 9\omega, 3w^2\}, \quad u, v \in \mathbf{Q}. \quad (12)$$

For $\alpha = 1$, [1, Lemma 6.4] further gives $w, u, v \in \mathbf{Z}$, with $\gcd(Z, w) = \gcd(u, v) = 1$.

Expanding (12) produces three genus-two curves. One has no $\mathbf{Q}_2$ -point because its primitive homogeneous sextic is always 5 modulo 8. A second has precisely the two rational points $(-2, \pm 540)$; its Jacobian has rank one and the rational points are exhausted by Chabauty's method. Substitution gives $r = -1/4$ and

$$\Phi(-1/4) = -\frac{491}{2^{14}},$$

which cannot be a reduced quotient of a fifth power by a seventh power.

3.2 The main genus-two curve

The remaining case gives the curve

$$C_1: \quad y^2 = x^6 - 6x^5 + 69x^4 - 38x^3 + 6x^2 - 240x + 1057. \quad (13)$$

Its Jacobian is torsion-free of rank three, and three explicit Mumford divisors form a $\mathbf{Z}$ -basis. A Mordell–Weil sieve in the sense of [13], modulo $1080J_1(\mathbf{Q})$, places every rational point in one of the six classes represented by

$$\infty_+, \infty_-, (1/2, \pm 245/8), (3, \pm 65).$$

Exact formal-group calculations on these classes imply, for every affine $(x, y) \in C_1(\mathbf{Q})$,

$$\text{ord}_3(x) \leq -3 \quad \text{or} \quad \text{ord}_5(y) \geq 1. \quad (14)$$

The point on (13) arising from (12) has

$$x = \frac{u}{v}, \quad y = \frac{(Z + w)^2 + 3w^2}{v^3}.$$

The case $v = 0$ is also visible on the projective model. Coprimality forces $u = \pm 1$, while the two descent equations give

$$Z^2 + wZ - 3w^2 = \pm 1, \quad -w(Z + w) = 0.$$

If $w = 0$, then $X = 0$; if $Z = -w$, then $-3w^2 = \pm 1$. Both are impossible under the hypotheses.

For $v \neq 0$, the second arm of (14) would imply $5 \mid (Z + w)^2 + 3w^2$, impossible for a primitive pair modulo 5. Hence $27 \mid v$. The second descent equation gives $27 \mid w(Z + w)$. Coprimality and (10) exclude $27 \mid w$, so $27 \mid Z + w$. Reducing the first descent equation modulo 3 now gives $3 \mid u$, contradicting $\gcd(u, v) = 1$.

Proposition 3.1. *The polynomial f_η has no rational root. Hence the sector $[1, 6]$ is empty.*

4 The quadratic-factor sector

Provenance. The reduction to the fibre product and its six rational points is Dahmen–Siksek, Proposition 5.1 and Lemma 5.2 [1, Section 5]. We include the construction because it is the complete bridge from a quadratic root to the final fifth-power test. The database-free recovery of $\mathbf{Q}(\sqrt{-35})$ from its local labels is the additional refinement used here.

Suppose that A_η has a quadratic direct factor L . Its real signature, ramification, and dyadic labels identify it without a field database.

Since f_η has exactly one real root, L is imaginary. At every prime outside $\{3, 5, 7\}$, the scaled local models are separable and every direct factor is unramified. At 2, the possible local degree patterns are $[1, 2, 4]$ and $[1, 3, 3]$. A quadratic global direct factor must therefore be the unique unramified quadratic component in the first pattern. Thus 2 is inert in L . The negative odd fundamental discriminants supported on $\{3, 5, 7\}$ and inert at 2 are -3 and -35 . Finally, the degree-two Newton-polygon component at 7 is ramified. Therefore

$$L = k = \mathbf{Q}(\omega). \quad (15)$$

Let u be a nonrational root of f_η in L and $v = \bar{u}$. Symmetric elimination gives a rational point on

$$D : \begin{cases} \beta^2 = \alpha^4 - 4\alpha^3 + 6\alpha^2 + 9, \\ \alpha(\alpha - 4) = -35\gamma^2. \end{cases} \quad (16)$$

Here

$$\alpha = \frac{(u + v)^2}{uv}.$$

Proposition 4.1. *The rational points on D are exactly*

$$(0, \pm 3, 0), \quad \left(\frac{35}{9}, \pm \frac{782}{81}, \pm \frac{1}{9} \right),$$

with independent signs in the second family.

Proof. Map D to the genus-two curve

$$-35y^2 = x(x - 4)(x^4 - 4x^3 + 6x^2 + 9). \quad (17)$$

The étale algebra of the right side is $\mathbf{Q} \times \mathbf{Q} \times L_4$, where

$$L_4 = \mathbf{Q}(\vartheta), \quad \vartheta^4 - 4\vartheta^3 + 6\vartheta^2 + 9 = 0.$$

An exact 2-cover descent [12] gives three fake-Selmer classes, whose components at 0, 4, and ϑ are

	0	4	ϑ
ξ_1	-3	1	$3(\vartheta - 4)$
ξ_2	-35	1	1
ξ_3	-35	1	-5.

The first class cannot lift to a nonendpoint rational point on D . Indeed, in this class $x = -3ta^2$ and $x - 4 = tb^2$, whereas the additional condition on D requires $x(x - 4)/(-35)$ to be a square. This would make $3/35$ a square, which is impossible already over $\mathbf{Q}_7$. The endpoint $x = 4$ does not lift either, since it would require $\beta^2 = 105$. The other two classes map to elliptic curves over L_4 .

Their ranks are respectively 3 and 2, their torsion is trivial, and elliptic Chabauty [11] retains only the rational x -coordinates

$$x = 0, \quad x = 35/9.$$

For the rank-three curve, completeness is certified by saturation at every prime dividing the Chabauty index

$$6592880678883323322765152793600 = 2^{10} 3^5 5^2 7^4 \cdot 11 \cdot 17 \cdot 19 \cdot 29 \cdot 31 \cdot 37 \\ \cdot 43 \cdot 47 \cdot 53 \cdot 61 \cdot 71 \cdot 83 \cdot 97.$$

Substitution into (16) gives precisely the six points stated. $\square$

If $\alpha = 0$, then $v/u = -1$. The root relation gives either $u = 0$ or $u = \pm\delta/5$. The nonzero alternatives satisfy

$$\Phi(u) = \frac{7^4}{5^2},$$

which is not a reduced quotient of a fifth power by a seventh power. If $\alpha = 35/9$, then

$$\frac{v}{u} = \frac{17 \pm \delta}{18}.$$

The root relation gives either a branch value or the rational number

$$\frac{495535757310206240323260222}{432344158742282808792052775},$$

whose coprime numerator and denominator factor as

$$495535757310206240323260222 = 2 \cdot 3^{16} 7^4 \cdot 17 \cdot 19^{10} \cdot 23, \\ 432344158742282808792052775 = 5^2 \cdot 29^7 \cdot 139^7.$$

The numerator is not a fifth power. This proves:

Proposition 4.2. *The sector $[2, 5]$ is empty.*

5 The cubic–quartic sector

Provenance. The elimination follows Dahmen–Siksek, Section 7, in particular Proposition 7.1 and Lemmas 7.2–7.7 [1, Section 7]. Our additional contribution is to certify the ramified local step inside the intrinsic integral formal neighbourhood, rather than treating model-dependent coordinates as an unexamined proxy for the formal group.

The quartic direct factor is totally imaginary and unramified outside $\{3, 5, 7\}$. Dahmen–Siksek consider the complete lists of nineteen cubic fields of signature $(1, 1)$ and seventeen quartic fields of signature $(0, 2)$ unramified outside this set. They compare each cubic–quartic product with the permitted local algebras at 3, 5, and 7. Their Proposition 4.3 leaves two products, whose cubic factors are $\mathbf{Q}(\sqrt[3]{5^\alpha 7})$ for $\alpha = 1, 2$, and whose quartic factor is the same field in both cases [1, Proposition 4.3].

The surviving field is

$$M = \mathbf{Q}[a]/(a^4 - a^3 + 3a^2 + 3a + 9) = k\left(\sqrt{\frac{7 + \delta}{2}}\right). \quad (18)$$

The associated quotient is the genus-two curve

$$C_2 : \quad V^2 = U(U - 4)(U^4 - 4U^3 + 6U^2 + 9) \quad (19)$$

over k . Write $J_2 = \text{Jac}(C_2)$.

5.1 The Mordell–Weil group and sieve

We use the Mordell–Weil group determination of [1, Lemma 7.3], which gives

$$J_2(k) = \langle D_0 \rangle_{10} \oplus \mathbf{Z}D_1 \oplus \mathbf{Z}D_2 \oplus \mathbf{Z}D_3. \quad (20)$$

For reproducibility, we also give a direct finite-field certificate for the remaining 2-saturation in this statement. Put $A = J_2(k)$, $H = \langle D_0, D_1, D_2, D_3 \rangle$, and $H_0 = \langle D_0, D_1, D_2, D'_3 \rangle$. The rational and twisted Mordell–Weil computations used in the cited lemma give $\text{rank } A = 3$; good reduction at 11 and 13, together with the displayed order of D_0 , gives $A_{\text{tors}} = \langle D_0 \rangle \simeq \mathbf{Z}/10\mathbf{Z}$. Moreover, $2A \subseteq H_0 \subseteq H$, the second inclusion following from the verified relation

$$D'_3 = 9D_0 + 3D_1 - D_2 - 2D_3.$$

It therefore remains only to show that the four displayed classes span $A/2A$.

At the good prime $\mathfrak{l} = (173, \delta - 22)$, the change of variables

$$u = -36/U, \quad v = 1296V/U^3$$

puts the curve into the odd monic form

$$v^2 = u^5 + 9u^4 + 864u^3 + 28512u^2 + 373248u + 1679616.$$

Its five roots modulo $\mathfrak{l}$, in the certified order, are 15, 90, 119, 122, 164. Applying the usual odd-degree $u - T$ Kummer map and then the five quadratic characters of $\mathbf{F}_{173}^\times$ to D_0, D_1, D_2, D_3 gives the matrix

$$\begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

The determinant of its first four rows is 1 over $\mathbf{F}_2$. Hence these four classes are independent in $A/2A$. Since $\dim_{\mathbf{F}_2} A/2A = 3 + 1 = 4$, they form a basis; as A/H was already killed by 2, this proves $A = H$. The archive contains both the short reconstruction and an independent evaluation of every Kummer entry.

A Mordell–Weil sieve [13] with modulus 2240 yields twelve points $P_1, \dots, P_{12}$ such that every $Q \in C_2(k)$ satisfies

$$[Q - P_i] \in 2240J_2(k) \quad (21)$$

for a unique i .

Let $\mathfrak{P}$ be the unique prime of k above 7, let $k_{\mathfrak{P}}$ be the completion, and put $R = \mathcal{O}_{k_{\mathfrak{P}}}$, $\mathfrak{m} = (\delta)$. Thus

$$v_{\mathfrak{P}}(\delta) = 1, \quad v_{\mathfrak{P}}(7) = 2.$$

Use Flynn’s integral $\mathbf{P}^{15}$ model of J_2 with origin $(1 : 0 : \dots : 0)$ and normalized coordinates $s_i = a_i/a_0$. On the origin neighbourhood $\mathcal{N}$, the projective-coordinate framework of [8] and Corollary 2.2 and Theorem 3.5 of [7] identify (s_1, s_2) as integral formal parameters and give an integral formal group law. Define

$$G[2] = \{A \in \mathcal{N} : s_1(A), s_2(A) \in \mathfrak{m}^2\}. \quad (22)$$

It is a subgroup, and the coordinate identities place every nonorigin normalized coordinate in $\mathfrak{m}^2$.

The exact origin-chart valuations of $2240D_1$, $2240D_2$, and $2240D_3$ in the adapted integral coordinates described in Appendix B are

$$\begin{array}{l|l} D_1 & 0, 2, 2, 4, 4, 4, 6, 6, 6, 6, 8, 8, 8, 8, 16 \\ D_2 & 0, 5, 3, 10, 8, 6, 15, 13, 11, 9, 20, 18, 16, 14, 12, 20 \\ D_3 & 0, 2, 2, 4, 4, 4, 6, 6, 6, 6, 8, 8, 8, 8, 16. \end{array} \quad (23)$$

The torsion generator is killed by 2240, so

$$2240J_2(k) \subset G[2]. \quad (24)$$

5.2 The twelve local classes

For each P_i , exact pullback calculations prove

$$[Q - P_i] \in G[2] \implies Q \equiv P_i \pmod{\mathfrak{P}^2}. \quad (25)$$

At each of five regular finite bases, the 98 depth-two congruence classes split into one base class and 97 rejected classes. At each of the five ramified finite bases, the complete tree is

n	tested	base	rejected	unresolved	49-digit lifts	admissible at $n + 1$
2	98	1	91	6	294	294
3	294	0	252	42	2058	686
4	686	0	588	98	4802	686
5	686	0	686	0	--	--.

The last two columns make the branching arithmetic explicit. Each unresolved pair modulo $\mathfrak{P}^n$ has $7^2 = 49$ two-coordinate digit lifts; the next row counts only those lifts that also satisfy the curve equation modulo $\mathfrak{P}^{n+1}$. At the ramified special-fibre point this number is not uniformly seven, which is why the tested-node column is not obtained by multiplying the unresolved column by a fixed branching factor. The singular special-fibre point $U \equiv 4 \pmod{\mathfrak{P}}$ is included in this calculation. At the two points at infinity, the opposite branch lies outside $\mathcal{N}$ and the identity branch has

$$s_1 = \pm \frac{1}{2}w + O(w^2),$$

with unit linear coefficient. Here a node is a full congruence class, and a rejection is a uniform valuation statement on that class; no finite sample is used as an exhaustion argument.

Combining (21), (24), and (25), every k -point lies in one of the twelve explicit $\mathfrak{P}^2$ -disks. Exact 7-adic and 23-adic conditions, followed where necessary by Coleman integration, show that no disk contains a solution-bearing point beyond its reference P_i . The reference points themselves either have parameter 0, 1, or ∞ , or fail the required rational admissibility condition; hence none yields a nonzero primitive solution.

Proposition 5.1. *The sector $[3, 4]$ is empty.*

6 The irreducible septic sector

Provenance. The seven-field enumeration is Putz's published theorem. The elimination of the six pure fields and the exceptional field is the new terminal part of the argument.

Suppose now that A_η is a field. It has degree 7, signature $(1, 3)$, is unramified outside $\{3, 5, 7\}$, and has the prescribed local algebras at 2, 3, 5, and 7 of [1, Corollary 3.4 and Propositions 3.5–3.7]. Putz's exhaustive Hunter enumeration [2, Theorem 3.50] gives exactly seven possibilities:

$$\mathbf{Q}\left(\sqrt[7]{3^a 5}\right) \quad (1 \leq a \leq 6), \quad \mathbf{Q}[T]/(T^7 - 483T^2 + 3955T - 3945). \quad (26)$$

We call the first six fields pure and the last field exceptional.

6.1 The pure fields

The Fano-resolvent construction and the rational-parameter theorem in this subsection supply the geometric obstruction that was absent at the published seven-field frontier.

The normal closure of a pure field has Frobenius group F_{42} and unique quadratic subfield

$$K_7 = \mathbf{Q}(s), \quad s^2 = -7.$$

6.1.1 The Fano-resolvent bridge

The passage from a pure septic field to a curve is as follows. Scale $V = 15T$ and write

$$h_t(V) = V^7 - 35V^6 + 315V^5 - 15^6t, \quad t = \eta = \frac{X^5}{Z^7}. \quad (27)$$

Let $v_1, \dots, v_7$ be its roots. A labelled Fano plane F on these seven roots has seven lines, each a three-element subset; define

$$M_F = \sum_{\{i,j,k\} \text{ a line of } F} v_i v_j v_k, \quad q_F = \frac{M_F}{225}. \quad (28)$$

There are thirty labelled Fano planes. They form two A_7 -orbits $\mathcal{F}_+$ and $\mathcal{F}_-$ of fifteen planes. Since

$$\text{disc}(h_t) = -7^7 15^{36} t^4 (t-1)^2,$$

the two orbit polynomials are conjugate over K_7 . After fixing the Vandermonde orientation, put

$$R_+(q, t) = \prod_{F \in \mathcal{F}_+} (q - q_F) = \frac{U(q, t) + sW(q, t)}{2}, \quad U, W \in \mathbf{Z}[q, t]. \quad (29)$$

The normalization of $R_+ = 0$ is denoted C_+ . On its fifteen sheets the branch permutations above $t = 0, 1, \infty$ have cycle structures

$$5^3, \quad 3^5, \quad 7^2 1.$$

Riemann–Hurwitz therefore gives $g(C_+) = 3$.

To see why this quotient detects the six pure fields, call $F \in \mathcal{F}_+$ and $G \in \mathcal{F}_-$ compatible when they have no line in common. We use unordered pairs $\{F, G\}$, since odd permutations exchange the two A_7 -orbits. Each F has exactly eight compatible G 's, giving 120 such pairs. The normalization of the corresponding cover is a curve $X_{42}/\mathbf{Q}$. The stabilizer of an unordered compatible pair in A_7 is the Frobenius group F_{21} , and its stabilizer in S_7 is F_{42} . Thus X_{42} has degree 120 over the t -line and, after base change to K_7 , the forgetful map

$$X_{42, K_7} \longrightarrow C_+ \quad (30)$$

has degree 8. The branch permutations on the 120 sheets have cycle structures 5^{24} , 3^{40} , and $7^{17} 1$, so $g(X_{42}) = 20$.

If (27) cuts out one of the pure fields $\mathbf{Q}(\sqrt[7]{3^a 5})$, its normal closure has group F_{42} . The fixed-field/resolvent correspondence supplies a rational point on X_{42} , hence a K_7 -point on C_+ through (30). Both maps preserve t . A nonzero solution has $t \notin \{0, 1, \infty\}$, so the resulting point is not in a branch fibre. This constructs the claimed bridge rather than merely inferring it from matching Galois groups.

The canonical model of C_+ is the explicit smooth plane quartic printed in Appendix A. Its smoothness proves again that its genus is 3. The inverse parameter is the exact rational function $t = -G_t/H_t$; all coefficients of the two quintics are printed in Table 2. The archive file `p7_f42_canonical_exact.json.gz` specifies the forward coordinates $N_j(q, t)$ for $j = 1, 2, 3$, and the inverse forms G_q, H_q . Exact reduction modulo $R_+(q, t)$ verifies

$$Q(N_1, N_2, N_3) = 0, \quad G_q(N) + qH_q(N) = 0, \quad G_t(N) + tH_t(N) = 0,$$

where Q is the printed quartic and the two inverse denominators are nonzero in the function field. Recovering both generators q and t proves that this map induces an isomorphism of function fields, hence of the smooth projective models, preserving the parameter.

Theorem 6.1 (rational-parameter theorem). *Let $J_+ = \text{Jac}(C_+)$. Then*

$$\{Q \in C_+(K_7) : t(Q) \in \mathbf{P}^1(\mathbf{Q})\} = \{P_1, P_2, P_3, P_4, P_5\}.$$

The corresponding parameter values are $0, 1, \infty, \infty, \infty$.

Proof. Put $D_i = [P_i - P_1]$. An exact 2-descent, implemented in the general genus-three framework of [9], and a rank calculation give

$$\dim_{\mathbf{F}_2} \text{Sel}_2(J_+/K_7) = 4, \quad \text{rank } J_+(K_7) = 4, \quad \text{Sha}(J_+/K_7)[2] = 0,$$

and D_2, D_3, D_4, D_5 form a basis modulo 2. Their subgroup therefore has odd index. The existence of the fifth-division class used below can be seen from the branch divisor. Write $A = (t)_0/5$, a K_7 -rational divisor of degree three, and set $B = [A - 3P_1]$. The pole divisor is $(t)_\infty = 7P_3 + 7P_4 + P_5$, so

$$5B = 7D_3 + 7D_4 + D_5.$$

Consequently the class $E = 3B - 4D_3 - 4D_4$ satisfies

$$5E = D_3 + D_4 + 3D_5. \tag{31}$$

We use (D_2, E, D_4, D_5) for the odd-primary part. Four independent local C_{25} rows have determinant 4 modulo 5, and $J_+(K_7)[5] = 0$; hence this subgroup is 5-saturated.

At the split primes

$$23, 43, 67, 71, 79, 107, 109, 113,$$

both reductions of the curve are enumerated, as are the images of every reduced point in the relevant local quotients of J_+ . The same global coefficient vector must occur at both places. In addition, rationality of $t(Q)$ imposes equality of the two projective reductions of t . The resulting finite sieve has the following exact counts:

stage	surviving coefficient classes
coupled 2-primary sieve modulo 16	592
5-primary prefilter modulo 25	1252
joint point fingerprints modulo 400	144

The final classes use exactly the five vectors modulo 25

$$(0, 0, 0, 0), (1, 0, 0, 0), (0, 5, 24, 22), (0, 0, 1, 0), (0, 0, 0, 1),$$

and their projection at the two primes above 23 consists precisely of the five ordered residue pairs of $P_1, \dots, P_5$.

The combined 23-adic logarithm matrix has size 6×4 and rank 4. Two independent annihilating differentials therefore vanish on all of $J_+(K_7)$. At the disks of P_2, P_3, P_4, P_5 , the Siksek tangent determinants modulo 23 are

$$12, \quad 16, \quad 4, \quad 19. \tag{32}$$

This is the number-field Chabauty criterion of [10]; the values are units, so each disk contains only its reference point among common Coleman zeros.

At P_1 , a normalized second-order system has exactly two simple reduced zeros,

$$(u, v) = (0, 0), \quad (14, 16),$$

with determinants 20 and 3. The first is P_1 . At the second, the two local parameter values modulo $23^8 = 78,310,985,281$ are

$$t_{s=4} = 59,233,664,629, \quad t_{s=19} = 24,239,267,738.$$

Their difference has exact 23-adic valuation 5, with first nonzero quotient digit 9. Hence this common Coleman zero does not have rational t . The five reference points are therefore the entire rational-parameter locus. $\square$

Each P_i in Theorem 6.1 lies above one of the branch values 0, 1, and ∞ . A nonzero primitive solution would give a point with rational parameter outside this set. Consequently none of the six pure fields occurs.

6.2 The exceptional field

The general modularity and conductor theorems used here are published inputs. The project-specific specialization, exhaustive packet and ray-character interfaces, and the terminal $q = 29$ comparison form the new finite elimination.

We use the Frey abelian variety of Pacetti and Villagra Torcomian [3] for

$$a^5 + b^p + c^3 = 0. \quad (33)$$

In our variables

$$(a, b, c) = (X, -Z, Y), \quad t_0 = -\frac{a^5}{c^3} = \frac{\eta}{\eta - 1}. \quad (34)$$

Put $F = \mathbf{Q}(\sqrt{5})$ and let $\bar{\rho}_7 : G_F \rightarrow \mathrm{GL}_2(\bar{\mathbf{F}}_7)$ be the residual representation of their plus motive at t_0 . The use of the modular method at the small exponent $p = 7$ rests on the following precise inputs, not on the headline large-exponent theorem of [3]. Theorem 2.4(1) of [3] states that for $r = 3$ and $q \geq 5$ the plus motive is modular for every rational specialization. Its determinant is the mod-7 cyclotomic character (equivalently, the compatible system has trivial Nebentypus), and Theorem 2.1(3)–(4) gives the needed ramification control and finiteness at every prime above 7. Here 7 is inert and unramified in F , so there is a single such prime.

Now assume that $\bar{\rho}_7$ is irreducible. Apply [3, Theorem 7.8] with $p = 7$ to the specialization (34). This theorem packages the modularity, local conductor calculation, and level-lowering argument for the plus motive: there is a parallel-weight-two Hilbert newform over F , with trivial Nebentypus, whose residual representation is $\bar{\rho}_7$ and whose level is one of

$$3^{\epsilon_3}(\sqrt{5})^{\epsilon_5}, \quad (\epsilon_3, \epsilon_5) \in \{2, 3\}^2. \quad (35)$$

The conclusion here is a newform conclusion, but the finite elimination below does not depend on reconstructing a new quotient. Let V_N be the full mod-7 cuspidal Brandt module at one of the four levels N in (35). The historical computation first imposed eighteen local trace-polynomial conditions, so its output must not be called the full ambient module. We instead verify the coverage of every condition.

Section 7.4 of [3] gives an exhaustive trichotomy at an auxiliary prime: ordinary specialization, one of the two degenerations $t = 0, \infty$, or the level-lowering specialization $t = 1$. From the complete upstream `Data.txt`, the certificate independently reduces all 6253 trace polynomials at 37 primes modulo 7, verifies all ordinary parameters and all five and three degenerate character values, performs the required relative-degree-two trace transport, and adjoins both signs of $N(\mathfrak{q}) + 1$. Seventeen of the eighteen historical filters are exactly the resulting squarefree allowed-trace polynomials. At $\ell = 131$ the historical degree-48 filter omitted the allowed level-lowering trace 1, since $-(131 + 1) \equiv 1 \pmod{7}$. We therefore replace only that condition, in memory, by $T^{49} - T$, which admits every $\mathbf{F}_{49}$ -valued trace.

Write $W_N \subseteq V_N$ for the intersection after these seventeen verified conditions and the conservative no-filter condition at 131. Characteristic-zero Jacquet–Langlands [6, Theorem 3.9 and §4] supplies the corresponding Hecke eigenvector in the definite quaternionic function model. The exact order and neighbour computations verify the maximal order, class number one, and the complete Hecke operators; all finite stabilizer orders divide 120, hence are prime to 7. The integral function lattice and its cuspidal mass condition are therefore preserved at primes above 7. Scale the eigenvector to be primitive in this local lattice and reduce. The resulting nonzero vector has the required residual eigenvalues. The local trichotomy places it in W_N . This proves coverage without degeneracy-map subtraction. A fresh simultaneous primary decomposition has

packet dimensions, with multiplicity, summing to $\dim W_N$ at every level. The prime-2 calculation gives the necessary residual trace -1 when X is even and 0 when Y is even. Retaining every packet for which that scalar is a root of the residual T_2 polynomial gives

	22	23	32	33
$\dim W_N$	8	38	28	98
all packets	4	14	12	24
X -even	1	2	2	5
Y -even	2	10	3	11.

Across all thirty-six retained packet occurrences, the set of residual T_q annihilators at $q \mid 29$ is exactly

$$T, \quad T^2, \quad T + 2, \quad T^2 + 5T + 2, \quad T^2 + 3T + 4.$$

The terminal list already contains all five, because $T + 2 = T - 5$ modulo 7; together with $T - 2$ it is the same six-factor list used in the resultant calculation below. Thus every residual representation supplied by level lowering survives the preceding filters and is covered before the terminal comparison. The old/new calculation remains only a diagnostic. We make no complete-newspace claim and require no mod-7 injectivity or semisimplicity statement for degeneracy maps.

Suppose instead that $\bar{\rho}_7$ is reducible. Its semisimplification has the determinant-compatible form

$$\bar{\rho}_7^{\text{ss}} = \psi \oplus \psi^{-1} \bar{\chi}_7. \quad (36)$$

At a finite prime $\lambda \nmid 7$, the cyclotomic character is unramified, so the two characters in (36) have the same Artin-conductor exponent. The conductor of the semisimplification is at most that of the representation. Thus the local conductor bounds at the primes above 3 and 5 give

$$2a_\lambda(\psi) = a_\lambda(\bar{\rho}_7^{\text{ss}}) \leq a_\lambda(\bar{\rho}_7) \leq 3, \quad a_\lambda(\psi) \leq 1.$$

At every other finite prime away from 7, the character ψ is unramified.

The prime $\mathfrak{p} = (7)$ is unramified and inert in F , with residue field $\mathbf{F}_{49}$. By the finite-flat character classification [5, Corollary 3.4.4], its restriction to inertia has the form

$$\psi|_{I_{\mathfrak{p}}} = \omega_2^{r_0 + 7r_1}, \quad r_0, r_1 \in \{0, 1\},$$

The character has level dividing two because it is a rank-one character of the full local Galois group. Here ω_2 is a fundamental character of level two, normalized so that $\omega_2^8 = \bar{\chi}_7|_{I_{\mathfrak{p}}}$. The two mixed possibilities must be excluded globally. Put $\varepsilon = (1 + \sqrt{5})/2$ and

$$u = \varepsilon^8 = 13 + 21\varepsilon.$$

The unit u is totally positive and satisfies

$$u \equiv 1 \pmod{3(\sqrt{5})}, \quad u \equiv -1 \pmod{7}.$$

Global reciprocity applied to u shows that the value of ψ on its local reciprocity image at $\mathfrak{p}$ is 1: at every other finite place this follows from the conductor bounds just proved, and at both real places it follows from total positivity. The displayed inertial character gives that value as $(-1)^{r_0 + r_1}$. Hence $r_0 = r_1$. If both are zero then ψ is unramified at $\mathfrak{p}$; if both are one then $\psi^{-1} \bar{\chi}_7$ is unramified there. Interchanging the two characters if necessary, we may therefore choose ψ such that

$$\text{cond}(\psi) \mid 3(\sqrt{5}). \quad (37)$$

Including both real places gives a modulus containing the conductor of ψ regardless of its signs. Its narrow ray group is $C_4 \times C_2$, so there are eight possible characters, all of which are included in the comparison.

At $q = 29$, take $\mathfrak{q} = (29, \sqrt{5} - 11)$; the conjugate prime gives the conjugate packet and the same common-root test. The exceptional septic polynomial is squarefree modulo 29 with factor degrees $[1, 3, 3]$. Among $\eta \in \mathbf{F}_{29} \setminus \{0, 1\}$, the polynomial $\Phi(T) - \eta$ has this pattern precisely for $\eta = 10, 14, 24, 28$.

The branch residues do not give this pattern. If $29 \mid X$, scale the five-root cluster at $T = 0$ using $\text{ord}_{29}(\eta) = 5 \text{ord}_{29}(X)$. Its separable reduction has degrees $[1, 2, 2]$, since $29 \equiv -1 \pmod{5}$; the complementary quadratic has degrees $[1, 1]$. If $29 \mid Y$, the corresponding three-root cluster at $T = 1$ has degrees $[1, 2]$, and the complementary quartic has degrees $[1, 1, 2]$. Both cases therefore have pattern $[1, 1, 1, 2, 2]$. If $29 \mid Z$, the scaled coordinate $U = ZT$ gives reduction $15U^7 - X^5$; its pattern is $[1, 1, 1, 1, 1, 1, 1]$ or $[7]$ because $\mathbf{F}_{29}$ contains the seventh roots of unity. Hensel's lemma applies to each scaled separable reduction. Thus only the four ordinary values remain. By (34) they correspond to $t_0 = 14, 10, 25, 15$. Direct point counting on

$$y^2 = 5x^6 - 12x^5 + 10t_0x^3 + t_0^2$$

gives the following polynomials over $\mathbf{F}_7$ satisfied by the two conjugate real-multiplication Frobenius traces. Explicitly, if $N_1 = \#C_{t_0}(\mathbf{F}_{29})$, $N_2 = \#C_{t_0}(\mathbf{F}_{29^2})$, $A = 30 - N_1$, and $B = (A^2 - (842 - N_2))/2$, the degree-four Frobenius polynomial is $X^4 - AX^3 + BX^2 - 29AX + 29^2$, and the two traces satisfy $T^2 - AT + (B - 58)$. The table records the latter polynomial modulo 7:

η	t_0	curve polynomial
10	14	$T^2 + 6T + 6$
14	10	$T^2 + 6T + 4$
24	25	$T^2 + 4$
28	15	$T^2 + 2T + 3$.

(38)

For clarity, the next list is an *annihilating family*; its members are not asserted to be irreducible or squarefree. For each of the four ambient cuspidal spaces, reduce the characteristic polynomial of the Hecke operator $T_{\mathfrak{q}}$ modulo 7 on every packet surviving the conservative T_2 condition. The preceding dimension accounting proves that every residual eigenvalue $a_{\mathfrak{q}}(f) \pmod{\mathfrak{p}}$, for every relevant newform and each coefficient prime $\mathfrak{p} \mid 7$ realizing the required Frey congruence, is a root of one of the first four or the sixth displayed factors. We retain the factors as they occur in the packet calculation, so they need not be squarefree or minimal. The reducible characters contribute the possible traces $\psi(\mathfrak{q}) + \psi(\mathfrak{q})^{-1}$. The resulting complete annihilating family is

$$T, \quad T^2, \quad T^2 + 5T + 2, \quad T^2 + 3T + 4, \quad T - 2, \quad T - 5. \quad (39)$$

Here T^2 and $T^2 + 3T + 4 = (T - 2)^2$ deliberately record multiplicity; for the resultant test they have the same roots as T and $T - 2$. For the specified prime $\mathfrak{q}$, the ray class has order two: a totally positive generator is $6 - \varepsilon$, whose square is congruent to the totally positive unit ε^2 modulo $3(\sqrt{5})$; its class is nontrivial, since modulo 3 the generator is $-\varepsilon = \varepsilon^5$, whereas every totally positive unit is an even power of ε , whose order modulo 3 is 8. Thus the reducible traces are 2 and -2 , covered by the displayed linear factors. The 24 pairwise resultants of (38) and (39) form the matrix

$$\begin{pmatrix} 6 & 1 & 5 & 1 & 1 & 5 \\ 4 & 2 & 3 & 1 & 6 & 3 \\ 4 & 2 & 6 & 1 & 1 & 1 \\ 3 & 2 & 6 & 2 & 4 & 3 \end{pmatrix} \pmod{7}. \quad (40)$$

Every entry is nonzero. Thus the exceptional field is incompatible with both the irreducible and reducible residual alternatives.

Proposition 6.2. *The sector [7] is empty.*

7 Proof of the main theorem

Proof of Theorem 1.1. Suppose that a nonzero primitive solution exists and reorder it as in (2). The descent algebra A_η belongs to exactly one of the four sectors in (7). Proposition 3.1 excludes [1, 6]. Proposition 4.2 excludes [2, 5]. Proposition 5.1 excludes [3, 4]. Proposition 6.2 excludes [7]. Every possible factorization is therefore impossible, a contradiction. $\square$

8 Computational certification

The computer-assisted portions of the proof are finite statements over $\mathbf{Q}$, number fields, finite fields, and truncated local rings. They are organized so that discovery is not used as evidence: every accepted output is checked against exact input data and against the mathematical predicate stated in the relevant proposition. The certificate archive serves two different purposes. It independently reconstructs finite calculations from the published part of the proof, and it supplies the load-bearing arithmetic for the new seven-field elimination. The provenance column below keeps those purposes distinct.

8.1 Exact arithmetic layer

The exact layer checks polynomial identities, resultants, discriminants, field embeddings, ideal-class representatives, divisor relations, point incidence, and rational substitutions. Integer and rational calculations are repeated without floating-point arithmetic. Manifests identify the input bytes and the corresponding outputs.

8.2 Finite exhaustion layer

Whenever a finite search is used as an exhaustion, the searched set is defined before enumeration. This applies to local factor patterns, Selmer classes, reduced curve points, Mordell–Weil coefficient classes, residue trees, ray characters, newform packets, and resultant comparisons. A residue-tree node denotes a complete congruence class. A rejected node is accompanied by a uniform congruence or valuation obstruction; it is not a sample point.

8.3 Local analytic layer

Formal-group and Coleman calculations record their working precision, unit pivots, ranks, determinants, and higher-precision replays. Divisions are made only by certified units or with their exact loss of precision recorded. Simple-root conclusions are accompanied by a nonzero reduced Jacobian determinant. Singular classes are subdivided or normalized until the stated conclusion follows.

8.4 Software and trust boundary

The calculations use PARI/GP [15] for exact number-field arithmetic, SageMath [16] for curve and finite-field calculations, and Magma [14] for descent, Mordell–Weil, saturation, newform, and Chabauty routines. Their role is the same as that of large integer arithmetic or a table of modular forms in a conventional proof: the inputs, outputs, and verification conditions are part of the mathematical record. The argument also imports the general theorems cited in the bibliography, rather than attempting to reprove them from first principles.

8.5 Companion certificate archive

The companion archive preserves the evidence in repository-relative paths and provides a single integrity and interface verifier. Running

`python3 -B scripts/verify_dependency_closed_release_v3.py`

from the archive root checks the declared release manifest, the 52 base evidence manifests, the exact 765-file repository closure, the explicit added-file and generator-input indexes, and consistency of the theorem-composition graph. The optional pair of flags `-run-sector-replays` `-run-foundational-replays` reruns both the ten sector entry points and the foundational reconstruction: the two independent five-place workers, their semantic comparison, the Selmer projector and rank-four deduction, both split-23 global logarithm matrices, the combined annihilator, and the exceptional local calculation. The distributed complete log was produced from a clean extraction in an isolated filesystem in which the original workspace path was absent. In the arXiv source bundle the release is the ancillary file `anc/PRIMITIVE_357_REPRODUCIBILITY_V3.tar.gz`; the accompanying `PROGRAMS_AND_CERTIFICATES.md` gives proposition-by-proposition commands and paths. The principal correspondence is:

mathematical assertion	provenance	certificate family
four exhaustive factor sectors Proposition 3.1	published; reconstructed published; reconstructed	signed global-algebra superselection rational-factor composition and its Mordell–Weil, sieve, and local packages
Proposition 4.2	published, with new intrinsic routing	database-free quadratic router and quadratic-factor reconstruction
Proposition 5.1	published, with new formal-group certification	quartic router, Mordell–Weil basis, sieve, and intrinsic formal-group packages
Theorem 6.1 and Appendix A exceptional-field elimination	new new, using published modular theorems	Fano-cover, canonical-quartic, rational-parameter, and split-prime packages Putz–PVT interface and finite $q = 29$ comparison packages
Proposition 6.2	new terminal composition	irreducible-sector composition package
Theorem 1.1	new global composition	complete four-sector proof graph and terminal certificate

The archive distinguishes three operations. The default verifier is a dependency-free check of identity, coverage, and exact finite interfaces. The supplied Python and PARI/GP programs rerun the open exact-arithmetic layers. SageMath and Magma programs rerun the curve, descent, modular-form, and Chabauty layers when those systems are available; authenticated logs are included for computations requiring the proprietary Magma runtime. Imported published theorems remain bibliographic dependencies and are not represented as newly proved computational claims.

9 Concluding perspective

The proof is not driven by the size of a putative solution. It represents that solution simultaneously by a descent algebra, local factor labels, divisor classes, finite quotient coordinates, and p -adic analytic data. Each representation imposes an exact compatibility condition. The global contradiction comes from the empty intersection of those conditions.

This organization is useful beyond the present signature. It separates three tasks that are often entangled: enumerating the possible algebraic states, transporting a state to a curve

where a finitely generated group acts, and testing the remaining local states with analytic functions. It also makes the proof modular: a general theorem can be replaced by an independent implementation without changing the surrounding argument, while the finite interfaces remain explicit.

A The plane quartic in the pure septic case

Let $K_7 = \mathbf{Q}(s)$ with $s^2 = -7$. In the monomial order

$$x^4, x^3y, x^3z, x^2y^2, x^2yz, x^2z^2, xy^3, xy^2z, xyz^2, xz^3, y^4, y^3z, y^2z^2, yz^3, z^4,$$

write the coefficient of each monomial as $(U + sW)/2$. The pairs (U, W) for C_+ are recorded below with the monomial repeated on every row, so that each coefficient can be quoted and checked independently of the surrounding order.

Table 1: Exact coefficients of the plane quartic C_+ . The coefficient of m is $(U + sW)/2$.

monomial m	U	W
x^4	526589143431946875/69798278	-114868376351859375/69798278
x^3y	-1271586888400500/3172649	-870445613016000/3172649
x^3z	72385949750625/34899139	-402884946883125/34899139
x^2y^2	-666163656476955/34899139	264021753009735/34899139
x^2yz	-74662865230275/34899139	45051674372775/34899139
x^2z^2	-20453292214425/69798278	1113321269025/69798278
xy^3	77961057423858/174495695	1554848756622/34899139
xy^2z	5974117172919/34899139	534487214709/34899139
xyz^2	251941629933/34899139	209684218779/34899139
xz^3	-13800316635/69798278	9656077725/69798278
y^4	-11467163171457/8724784750	-8293918213539/8724784750
y^3z	-192270890133/174495695	-111061125567/174495695
y^2z^2	-4669897617/31726490	-23997395583/158632450
yz^3	4101243048/174495695	-629073270/34899139
z^4	2	0

The equation and its three first partial derivatives have no common projective zero over K_7 ; this is checked by exact elimination. As an independent good-reduction certificate, both reductions at the primes above 23 are smooth plane quartics. Thus C_+ is smooth, nonhyperelliptic, and has genus 3. The two reductions and their common zeta numerator are recorded in Appendix C. The five points in Theorem 6.1 are

$$\begin{aligned} P_1 &= [-113 - 139s : 32925 + 11475s : -138600 - 57960s], \\ P_2 &= [1109 - 305s : 4325 - 19625s : 163800 - 37800s], \\ P_3 &= [-1 + s : 50 : 0], \\ P_4 &= [5 + s : 400 : 0], \\ P_5 &= [-157167 - 154941s : -82225175 + 2993875s : 186832800]. \end{aligned}$$

The homogeneous forms G_t and H_t have degree 5. For each monomial $m = x^a y^b z^c$ of degree 5, write

$$[m]G_t = \frac{U_{G,m} + sW_{G,m}}{2}, \quad [m]H_t = \frac{U_{H,m} + sW_{H,m}}{2}.$$

Table 2 gives the four rational components for every monomial; a component is displayed as separate numerator and positive denominator. These data define the parameter completely, with no external coefficient file required.

Table 2: Exact coefficient ledger for the inverse parameter $t = -G_t/H_t$. For each monomial and each form, the two successive rows are the rational components U and W of the coefficient $(U + sW)/2$.

monomial	component	rational value
G_t		
x^5	U	$\frac{8724124647223604676837903272778053868067651476251979316420946130788203125}{1262470913738019638514233667224982865889496755254608725372117056}$
x^5	W	$\frac{104539127651167761178849503246013584932810386714540106341619861276432421875}{34086714670926530239884309015074537379016412391874435585047160512}$
x^4y	U	$\frac{160516210802709714993563463849330968569737125845236059453160939760278125}{516465373801917124846731954773856626954794127149612660379502432}$
x^4y	W	$\frac{-1516029210727323321897909052657899390133429038437597834516845689254996875}{4648188364217254123620587592964709642593147144346513943415521888}$
x^4z	U	$\frac{-3431261016436648118361864834749807161605723145608327043665883055988296875}{536865756067092851278177866987423963719508495172022360464492778064}$
x^4z	W	$\frac{-14405986606016247091870808274044492866667442818955446375070000269331640625}{536865756067092851278177866987423963719508495172022360464492778064}$
x^3y^2	U	$\frac{-156765051716182227120181782804196796183006994771851219874584685103262375}{6391259000798724419978307940326475758565577323476456672196342596}$
x^3y^2	W	$\frac{-2797425838906852452662912431708042341593603601309085420184135364665875}{3195629500399362209989153970163237879282788661738228336098171298}$

Table 2 (continued)

monomial	component	rational value
x^3yz	U	$-\frac{2971418364998561881697119480380162120182637893006270782575715726325204375}{805298634100639276917266800481135945579262742758033540696739167096}$
x^3yz	W	$\frac{581935828601695609265701070325316097436069202977166807199027638486990625}{805298634100639276917266800481135945579262742758033540696739167096}$
x^3z^2	U	$-\frac{1101411663992169427898989261604416930376364487448681819065769508346053125}{11274180877408949876841735206735903238109678398612469569754348339344}$
x^3z^2	W	$-\frac{107499314624920129138097362917410399320732876511214060737904168396284375}{3758060292469649958947245068911967746036559466204156523251449446448}$
x^2y^3	U	$\frac{456938855157174881485528038353970608056813975973606988967508412987185}{4260839333865816279985538626884317172377051548984304448130895064}$
x^2y^3	W	$\frac{1661711353625849642579873501624741967233922652605715624836234484805605}{12782518001597448839956615880652951517131154646952913344392685192}$
x^2y^2z	U	$\frac{8474129667022123046619324483315222687658910122135070504455178101736425}{115042662014377039559609542925876563654180391822576220099534166728}$
x^2y^2z	W	$\frac{10719261028668655910991493734095990759234174142059301575470250588670875}{268432878033546425639088933493711981859754247586011180232246389032}$

Table 2 (continued)

monomial	component	rational value
x^2yz^2	U	$\frac{4788522504016908727437175426249647578416822156275084241334728620014875}{16911271316113424815262602810103854857164517597918704354631522509016}$
x^2yz^2	W	$\frac{5591416257739747140557321332572949584770401809552651792943380516721625}{1879030146234824979473622534455983873018279733102078261625724723224}$
x^2z^3	U	$\frac{-673613683718293366731089090657452594859133231945568100511338992953086875}{4261640371660583053446175908146171424005458434675513497367143672272032}$
x^2z^3	W	$\frac{727154975549102369129278091212463518913101596132005437190890995061885625}{4261640371660583053446175908146171424005458434675513497367143672272032}$
xy^4	U	$\frac{181934888585282020021801499997708888564901966896834717514388222336671}{127825180015974488399566158806529515171311546469529133443926851920}$
xy^4	W	$\frac{-66576524945702734300367034318330080309880903641645008587735355126139}{127825180015974488399566158806529515171311546469529133443926851920}$
xy^3z	U	$\frac{55752557283144284245053396819690398445556264921670780430266851605377}{134216439016773212819544466746855990929877123793005590116123194516}$
xy^3z	W	$\frac{-154753525348486993811829134912142488930864130682803774941942077141595}{402649317050319638458633400240567972789631371379016770348369583548}$

Table 2 (continued)

monomial	component	rational value
xy^2z^2	U	$\frac{45466049893188675406917805349188754066362340943964804016941934609315}{1537388301464856801387509346373077714287683417992609486784683864456}$
xy^2z^2	W	$\frac{-90644573939421031699724276084922394905211688255008129426169575178615}{1537388301464856801387509346373077714287683417992609486784683864456}$
xyz^3	U	$\frac{86176315804162437100883836603358200621380221474103257474111818725307525}{6392460557490874580169263862219257136008187652013270246050715508408048}$
xyz^3	W	$\frac{-14901625678223850735415766370606467751263679923368054621027598776871975}{6392460557490874580169263862219257136008187652013270246050715508408048}$
xz^4	U	0
xz^4	W	0
y^5	U	$\frac{-50208669906374986341132909015146131126007030919811841716909}{18313514842869689191983526987088351258653049645369350436400}$
y^5	W	$\frac{-118965057882822154213596381597899423461490949136300188722323}{91567574214348445959917634935441756293265248226846752182000}$

Table 2 (continued)

monomial	component	rational value
$y^4 z$	U	$-\frac{317869767610131557388806079392661467805959663406188675965981}{164821633585827202727851742883795161327877446808324153927600}$
$y^4 z$	W	$-\frac{20431957573101268107747459729242315992574863384255199500651}{32964326717165440545570348576759032265575489361664830785520}$
$y^3 z^2$	U	$-\frac{90218411306089523014211143406416603450262335284699082090763}{346125430530237125728488660055969838788542638297480723247960}$
$y^3 z^2$	W	$-\frac{6384772760573948911903427720751451666967418352649934952949}{69225086106047425145697732011193967757708527659496144649592}$
$y^2 z^3$	U	$\frac{34563607476891939880909708193414566015508357013210019043845}{13083541274042963352536871350115659906206911727644771338772888}$
$y^2 z^3$	W	$-\frac{150431184828356847118153754501573195221720737251858300004385}{13083541274042963352536871350115659906206911727644771338772888}$
yz^4	U	0
yz^4	W	0
z^5	U	0
z^5	W	0

Table 2 (continued)

monomial	component	rational value
H_t		
x^5	U	$\frac{417540987588276327046139642388996781228921135238823876465270116076953125}{55097060351713141551537137416607549642806701064452212691347781}$
x^5	W	$\frac{271400501016398695076663568050301774206517949977640898218621509239453125}{55097060351713141551537137416607549642806701064452212691347781}$
x^4y	U	$\frac{40800518279237110617651643358322019052974596058651349157620400597734375}{40070589346700466582936099939350945194768509865056154684616568}$
x^4y	W	$\frac{-10475518577744892148533246091047638248959642317009125512752949147203125}{40070589346700466582936099939350945194768509865056154684616568}$
x^4z	U	$\frac{295466106564310819299307055245131237369729837129047525997756150038046875}{13884459208631711670987358628985102509987288668241957598219640812}$
x^4z	W	$\frac{-97401726931434571033238656526295780061095710676931455291585333023515625}{13884459208631711670987358628985102509987288668241957598219640812}$
x^3y^2	U	$\frac{-2408541684482196326060738206798781548127364231879824910739247944739375}{73462747135617522068716183222143399523742268085936283588463708}$
x^3y^2	W	$\frac{-1401617439695752490670177104164771958374512006836961227347342807905625}{220388241406852566206148549666430198571226804257808850765391124}$

Table 2 (continued)

monomial	component	rational value
x^3yz	U	$-\frac{88727646196045391066194147279337983419623095543345635287685584174321875}{13884459208631711670987358628985102509987288668241957598219640812}$
x^3yz	W	$-\frac{294670882430653452222330070241389024327059653868933829299253113096875}{661164724220557698618445648999290595713680412773426552296173372}$
x^3z^2	U	$-\frac{143378643984727743099590687262597995420899449430994725195546663512734375}{1166294573525063780362938124834748610838932248132324438250449828208}$
x^3z^2	W	$-\frac{18365708170095511831980056438251994847190327220498321285191544780928125}{166613510503580540051848303547821230119847464018903491178635689744}$
x^2y^3	U	$\frac{8623734371967307823196434999403662938951554222083721534345295726825}{110194120703426283103074274833215099285613402128904425382695562}$
x^2y^3	W	$\frac{18881132466481791978253130146174121549567643122623579546749820507925}{110194120703426283103074274833215099285613402128904425382695562}$
x^2y^2z	U	$\frac{28562333150244501488893362369966137889574552762163770883392632257375}{661164724220557698618445648999290595713680412773426552296173372}$
x^2y^2z	W	$\frac{127302955155020767473662087940112338634670901473338060096039562505625}{1983494172661673095855336946997871787141041238320279656888520116}$

Table 2 (continued)

monomial	component	rational value
x^2yz^2	U	$\frac{111047168050090146883232536414177868796628159548247958136634257244625}{583147286762531890181469062417374305419466124066162219125224914104}$
x^2yz^2	W	$\frac{808931698949954129585743924035840186651521792034283998632202423247375}{194382428920843963393823020805791435139822041355387406375074971368}$
x^2z^3	U	$\frac{56671479059928976818861058588328919604601251570358478252116683982508125}{440859348792474108977190611187534974897116389794018637658670035062624}$
x^2z^3	W	$\frac{71221107904627237798651209022651158397866104168849046635811093612975625}{440859348792474108977190611187534974897116389794018637658670035062624}$
xy^4	U	$\frac{15345599112192921388150576223890339236281253816603221545634911025}{18365686783904380517179045805535849880935567021484070897115927}$
xy^4	W	$\frac{-32203376966478498296624374226292138386494075379549015420791430453}{55097060351713141551537137416607549642806701064452212691347781}$
xy^3z	U	$\frac{3144967508609794715796497440988774261265066246157112647816548077300}{3471114802157927917746839657246275627496822167060489399554910203}$
xy^3z	W	$\frac{-1516919058667068918402436722904874293272390262093914022875222741380}{3471114802157927917746839657246275627496822167060489399554910203}$

Table 2 (continued)

monomial	component	rational value
xy^2z^2	U	$\frac{1438698269678151371696913059361694896892235546906064110725007289965}{6626673713210589661153057527470162561584842318933661580968464933}$
xy^2z^2	W	$\frac{-80096480704524798441566620354536123544154597802973268379738357665}{946667673315798523021865361067166080226406045561951654424066419}$
xyz^3	U	$\frac{3102145252841255952780604187946926413931338657791125868394786111962775}{220429674396237054488595305593767487448558194897009318829335017531312}$
xyz^3	W	$\frac{63124143098168155026060265986978624933421514365737963701974316645275}{220429674396237054488595305593767487448558194897009318829335017531312}$
xz^4	U	0
xz^4	W	0
y^5	U	$\frac{186871683168118003488935611555116889880084010}{32767303919729427758399266748957493087161279}$
y^5	W	$\frac{25444070085364078630942054472628302788437558}{32767303919729427758399266748957493087161279}$

Table 2 (continued)

monomial	component	rational value
$y^4 z$	U	2
$y^4 z$	W	0
$y^3 z^2$	U	0
$y^3 z^2$	W	0
$y^2 z^3$	U	0
$y^2 z^3$	W	0
yz^4	U	0
yz^4	W	0
z^5	U	0
z^5	W	0

The exact canonical-map identity on C_+ is

$$G_t + tH_t = 0, \quad (41)$$

and therefore $t = -G_t/H_t$ away from the common projective base locus. The certificate verifies (41) in the homogeneous coordinate ring and checks the base points in the three branch fibres. Direct substitution gives $t(P_1) = 0$, $t(P_2) = 1$, and $t(P_3) = t(P_4) = t(P_5) = \infty$. Local expansions on the smooth quartic give

$$\text{ord}_{P_1}(t) = 5, \quad \text{ord}_{P_2}(t - 1) = 3, \quad \text{ord}_{P_3}(1/t) = \text{ord}_{P_4}(1/t) = 7, \quad \text{ord}_{P_5}(1/t) = 1.$$

In particular, P_5 is an unramified point above the branch value ∞ .

B Integral coordinates for the genus-two formal group

On the divisor-pair chart, write $\alpha_1 = x_1 + x_2$ and $\alpha_2 = x_1x_2$. The unnormalized rational coordinate functions in Flynn's projective embedding include

$$a_{15} = 1, \quad a_{14} = \alpha_1, \quad a_{13} = \alpha_2, \quad a_{12} = \alpha_1^2 - 2\alpha_2, \quad a_{11} = \alpha_1\alpha_2, \quad a_{10} = \alpha_2^2.$$

These formulas use the representative with $a_{15} = 1$; they do not normalize a_0 to 1. Near the projective origin, the normalized coordinates are instead $s_i = a_i/a_0$. In particular, $a_0a_{15} = a_5^2$, so $s_{15} = s_5^2$ and s_{15} vanishes at the origin.

The computations use the following integral coordinate block:

$$b_i = a_i \quad (0 \leq i \leq 11), \quad b_{12} = a_{13}, \quad b_{13} = a_{14}, \quad b_{14} = a_{15}, \quad b_{15} = a_{12} - 2a_{13}.$$

Its inverse is $a_{12} = b_{15} + 2b_{12}$, with all other displayed coordinates recovered directly. Hence this is an integral unimodular change of basis. Membership in the depth-two origin neighbourhood is therefore equivalent in the a - and b -coordinates. The valuation rows in (23) are stated in the b -coordinates.

C The split-prime data at 23 for the rational-parameter theorem

The prime 23 splits in K_7 as the two embeddings $s \mapsto 4$ and $s \mapsto 19$. Both reductions of C_+ have 20 rational points. Their common local numerator of the zeta function is

$$1 - 4T + 4T^2 + 56T^3 + 92T^4 - 2116T^5 + 12167T^6,$$

so

$$\#J_+(\mathbf{F}_{23}) = 10200, \quad 23 \nmid 10200.$$

The ordered reductions of the five points are

	$s = 4$	$s = 19$
P_1	(1, 21, 1)	(1, 11, 15)
P_2	(1, 0, 22)	(1, 8, 14)
P_3	(1, 9, 0)	(1, 13, 0)
P_4	(1, 1, 0)	(1, 9, 0)
P_5	(1, 11, 21)	(1, 6, 9).

For the regular differential basis η_x, η_y, η_z , the corresponding evaluation vectors are

	$s = 4$	$s = 19$
P_1	(3, 17, 3)	(11, 6, 4)
P_2	(4, 0, 19)	(5, 17, 1)
P_3	(4, 13, 0)	(2, 3, 0)
P_4	(12, 12, 0)	(22, 14, 0)
P_5	(7, 8, 9)	(10, 14, 21).

These data independently reproduce the tangent determinants in (32) once the two annihilator rows are applied.

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